Updating parameters

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

Before you start

For x = 2, target 3, w = 1 and b = 0, what is the squared-loss gradient in order [w, b]?

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Apply a learning rate to an ordered gradient. Try the recall before opening notes.

Your learning target

I can apply a **learning rate** to an ordered gradient.

I can update all parameters from the same saved state.

I can verify the new loss and identify **overshoot**ing.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Gradients with a numerical check. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Loss functions. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Composing layers. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Apply a **learning rate** to an ordered gradient

Update all parameters from the same saved state

Verify the new loss and identify **overshoot**ing

Retrieve what you already know

> For x = 2, target 3, w = 1 and b = 0, what is the squared-loss gradient in order [w, b]?

My starting explanation

This week's end goal

By the end, I can explain updating parameters using a traced example and a stated limitation.

Notice the evidence

> Investigate the supplied Updating parameters case. Separate observations, given facts and assumptions before choosing an explanation.

Gradient descent uses new parameter = old parameter - **learning rate** times its gradient entry. The minus sign chooses the locally downhill direction for a positive **learning rate**.

A negative gradient therefore increases the corresponding parameter.

Keep the Week 10 example: x = 2, target 3, y = 2w + b and L = (y - 3) squared. At [w, b] = [1, 0], the loss is 1 and gradient is [-4, -2].

With **learning rate** 0.1, new w = 1 - 0.1 times -4 = 1.4 and new b = 0 - 0.1 times -2 = 0.2.

Quantity	Before update	After update
Parameters [w, b]	[1, 0]	[1.4, 0.2]
Prediction	2	3
Squared loss	1	0

This particular step reaches zero loss on one example. It does not establish a unique parameter solution, successful learning on other examples or a generally ideal **learning rate**.

> **List the given facts and one fact that is not supplied.**

See how it works

> Follow the mechanism for Updating parameters. Connect each labelled input, deciding step and result.

```
def step(w, b, rate):
    residual = 2 * w + b - 3
    dw, db = 4 * residual, 2 * residual
    new_w = w - rate * dw
    new_b = b - rate * db
    new_loss = (2 * new_w + new_b - 3) ** 2
    return new_w, new_b, new_loss

for rate in [0.0, 0.05, 0.1, 0.3]:
    print(rate, tuple(round(v, 6)
                      for v in step(1, 0, rate)))
```

Predict the rows: unchanged parameters and loss 1 at rate 0; [1.2, 0.1] and loss 0.25 at rate 0.05; [1.4, 0.2] and loss 0 at rate 0.1; [2.2, 0.6] and loss 4 at rate 0.3.

The helper performs one step on this fixed example; it assumes numeric **input**s and does not choose a **learning rate** for you.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

> Compare the supplied cases for Updating parameters. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

For the lesson start [1, 0], apply rate 0.3 and calculate **prediction** and loss.

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change one thing

Change: Set **learning rate** 0.2 from the same starting point.

Predicted result

Prediction: New w=2.6, b=1.8, y=7, squared loss 4: no improvement.

Why the result follows

Reason: The larger update jumps to an equally distant point on the opposite side of the target.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

Use the vocabulary of Updating parameters to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Learning rate	A multiplier that controls the size of a gradient update
Gradient descent	Updating parameters in the direction opposite the gradient
Simultaneous update	Computing all new parameter values from the same old state
Overshoot	A step that moves too far and can increase the objective

Use two terms to explain a step in this week's example.

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Updating parameters, predict a step, then compare with its explanation.

Use one gradient for one parameter snapshot

STAGE	SAVED VALUES	ACTION
Before the step	w is 1 and b is 0	Compute gradient [-4, -2]
Apply rate 0.1	Use both old parameters	Produce [1.4, 0.2]
After the step	Use both new parameters	Recompute loss 0

Read top to bottom. All gradient entries come from the same old model. The final row evaluates the fully updated model rather than mixing old and new coordinates.

Starting again from [1, 0], use learning rate 0.05. Show both new parameters, the new prediction and the new loss.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Updating parameters

Use one explicit gradient update and then recompute the loss.

The facts we will use

Supplied facts

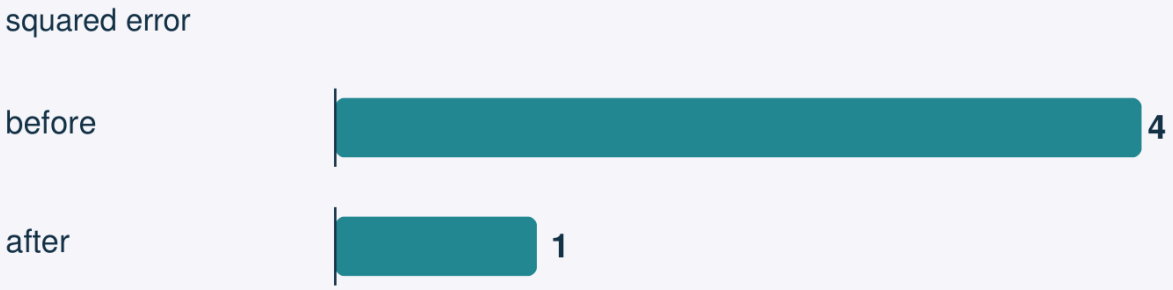
Model y=2w+b, target 5. Start w=1, b=1; **learning rate** 0.05. Initial y=3, error y−target=−2, squared loss 4.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/11

Loss before and after rate 0.05 update



Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/11

Updating parameters

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Compute gradient

Gradient for [w, b]=[2 x (-2) x 2, 2 x (-2)]=[-8,-4].



2. Update simultaneously

New w=1-0.05 x (-8)=1.4; new b=1.2.



3. Check new loss

New y=4, residual -1, loss 1: lower than 4.

CHANGE ONE THING

Set learning rate 0.2 from the same starting point.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Compute gradient	Gradient for [w, b]=[2×(-2)×2, 2×(-2)]=[-8,-4].
2. Update simultaneously	New w=1-0.05×(-8)=1.4; new b=1.2.
3. Check new loss	New y=4, residual -1, loss 1: lower than 4.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Gradient descent subtracts **learning rate** times gradient. A negative gradient makes this subtraction increase the parameter. Compute both gradients at the original [w,b] before updating either one.

The small step brings the **prediction** closer to five, reducing squared error. A larger step can cross the target and undo that gain, so the new **prediction** and loss must be recalculated.

What this example does not establish

Keep the stated limit

One successful toy update is not **evidence** of convergence for a real network or generalisation to unseen data.

Keep the stated limit

Read the labelled picture: Why must new loss be recomputed rather than assumed smaller? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: Set **learning rate** 0.2 from the same starting point. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Updating parameters claim and explain a repair using the supplied evidence.

Check the claim against the facts

Suppose a program first changes w to 1.4, then recomputes the bias gradient using new w and old b. Its residual is now -0.2, so its bias gradient is -0.4 and new b becomes 0.04. **Prediction** becomes 2.84 and loss 0.0256.

That is a different procedure from the declared simultaneous step, even though it also lowered this loss.

An excessive rate can increase loss. At rate 0.3, both coordinates move in the locally suggested direction but **prediction** jumps to 5 and loss rises from 1 to 4. Always evaluate and record the result.

Later lessons compare learning-rate behaviour across repeated updates.

Locate the earliest unsupported step

Explain why recomputing the bias gradient after changing w is not the **simultaneous update** specified here.

State one repair and the checks you would rerun.

STUDENT LESSON · 8 OF 18

Practise together

> Practise Updating parameters with support, then explain the deciding step in your own words.

Starting with parameter a = 2, gradient -3 and **learning rate** 0.2, calculate the new a.

Starting again from [1, 0], use learning rate 0.05. Show both new parameters, the new prediction and the new loss.

Explain your prediction to a partner before comparing results.

Practise with less help: Detect overshooting

Use workbook W2 for the connected example “Updating parameters”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Updating parameters. Proposed explanation: “Subtracting a negative gradient decreases the parameter”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Why must new loss be recomputed rather than assumed smaller? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

Run the investigation

> Investigate Updating parameters. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

For the lesson start [1, 0], apply rate 0.3 and calculate **prediction** and loss.

```
def step(w, b, rate):
    residual = 2 * w + b - 3
    dw, db = 4 * residual, 2 * residual
    new_w = w - rate * dw
    new_b = b - rate * db
    new_loss = (2 * new_w + new_b - 3) ** 2
    return new_w, new_b, new_loss

for rate in [0.0, 0.05, 0.1, 0.3]:
    print(rate, tuple(round(v, 6)
                      for v in step(1, 0, rate)))
```

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Updating parameters. Explain why each step is justified.

Explain → apply → challenge

Starting with parameter a = 2, gradient -3 and **learning rate** 0.2, calculate the new a.

Explain why recomputing the bias gradient after changing w is not the **simultaneous update** specified here.

> At a point with gradient [0, 0], what does this update do? Does that alone prove a global minimum on any loss surface?

> **Explain which evidence supports your conclusion and which claim remains unproved.**

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Image-pattern Network Investigation. Keep an annotated decision and the evidence supporting it.

Add a labelled updating parameters **evidence** sheet to Image-pattern Network Investigation. Show how you can apply a **learning rate** to an ordered gradient.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

A lower loss after a step supports progress on the objective and examples used for that step. It does not show that a model is safe, calibrated or better on unseen data.

Keep the data split fixed and never use final-test labels to compute **training** gradients.

A teammate says any update that lowers loss is a correct implementation of simultaneous gradient descent. Use the mixed-state example to evaluate that claim.

Next week repeats the forward, loss, gradient and update cycle in a small network with a clearly stated stopping budget.

At a point with gradient [0, 0], what does this update do? Does that alone prove a global minimum on any loss surface?

My limited claim and a useful next check

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Updating parameters case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

Use one gradient for one parameter snapshot

STAGE	SAVED VALUES	ACTION
Before the step	w is 1 and b is 0	Compute gradient [-4, -2]
Apply rate 0.1	Use both old parameters	Produce [1.4, 0.2]
After the step	Use both new parameters	Recompute loss 0

Read top to bottom. All gradient entries come from the same old model. The final row evaluates the fully updated model rather than mixing old and new coordinates.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Update a new model independently

Use x = 1, target 4, w = 1, b = 1, squared loss and **learning rate** 0.1. Compute **prediction**, residual, gradient [2 times residual times x, 2 times residual], new parameters and new loss.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

Use x = 1, target 4, w = 1, b = 1, squared loss and **learning rate** 0.1. Compute **prediction**, residual, gradient [2 times residual times x, 2 times residual], new parameters and new loss.

> **My independent procedure, result and limitation**

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Updating parameters, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Use one explicit gradient update and then recompute the loss.

By the end, I can	How I will show it
Apply a learning rate to an ordered gradient	Explain the deciding step, show a trace or test, and state one limit.
Update all parameters from the same saved state	Explain the deciding step, show a trace or test, and state one limit.
Verify the new loss and identify overshooting	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Updating parameters without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Apply a **learning rate** to an ordered gradient

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Use one explicit gradient update and then recompute the loss.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? New y=4, residual -1, loss 1: lower than 4.

1. Check new loss

2. Compute gradient

3. Update simultaneously

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Check new loss New y=4, residual -1, loss 1: lower than 4. **Gradient descent** subtracts **learning rate** times gradient.

A negative gradient makes this subtraction increase the parameter. Compute both gradients at the original [w,b] before updating either one. The small step brings the **prediction** closer to five, reducing squared error.

A larger step can cross the target and undo that gain, so the new **prediction** and loss must be recalculated.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Set **learning rate** 0.2 from the same starting point. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. New w=2.6, b=1.8, y=7, squared loss 4: no improvement.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: New w=2.6, b=1.8, y=7, squared loss 4: no improvement. The larger update jumps to an equally distant point on the opposite side of the target.

Repair a missing building block

Review Gradients with a numerical check (ai-5/10). Goal: Estimate one partial derivative while other parameters stay fixed. Short practice: Why should the numerical checks perturb one coordinate at a time?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Loss functions (ai-5/8). Goal: Calculate per-example squared error and mean squared error. Short practice: Why must the report name the chosen loss?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Composing layers (ai-5/5). Goal: Trace how one layer supplies the next layer **input**s. Short practice: What intermediate value explains the changed **output**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

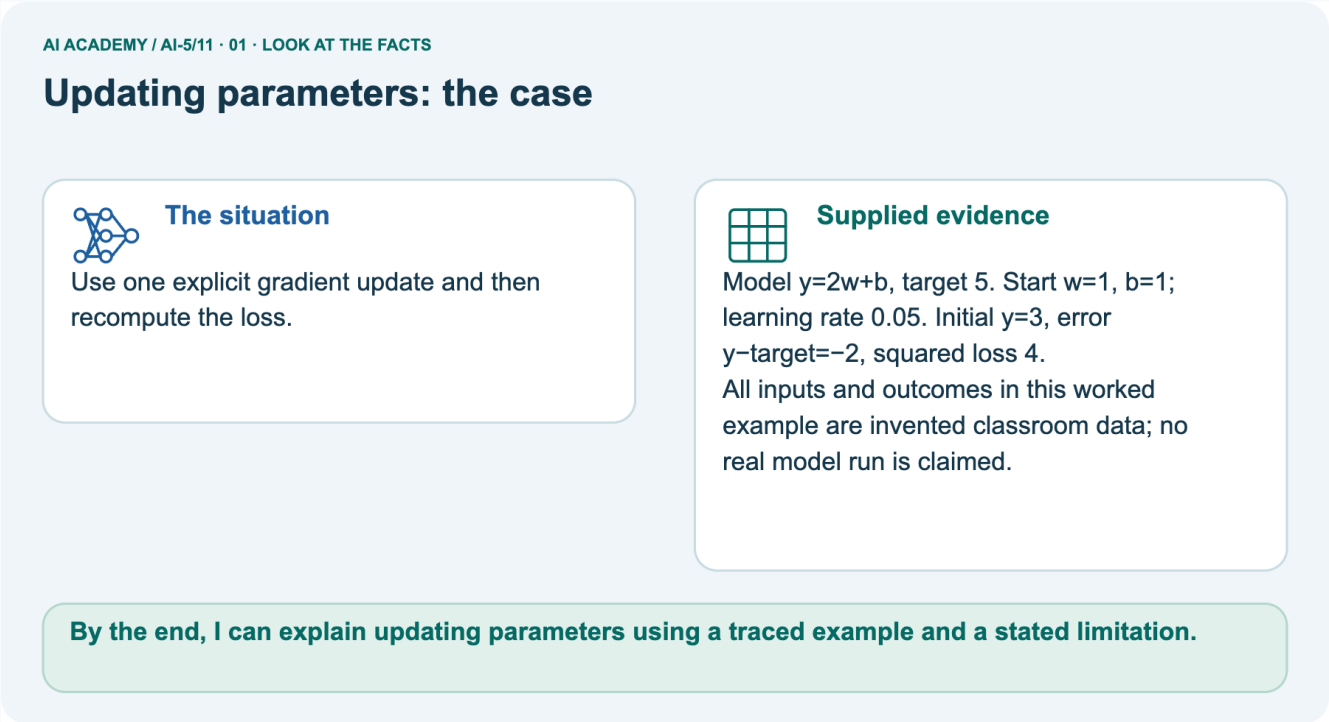
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Updating parameters graphic using its labels, deciding evidence and relevant limitation.

01. Updating parameters: the case



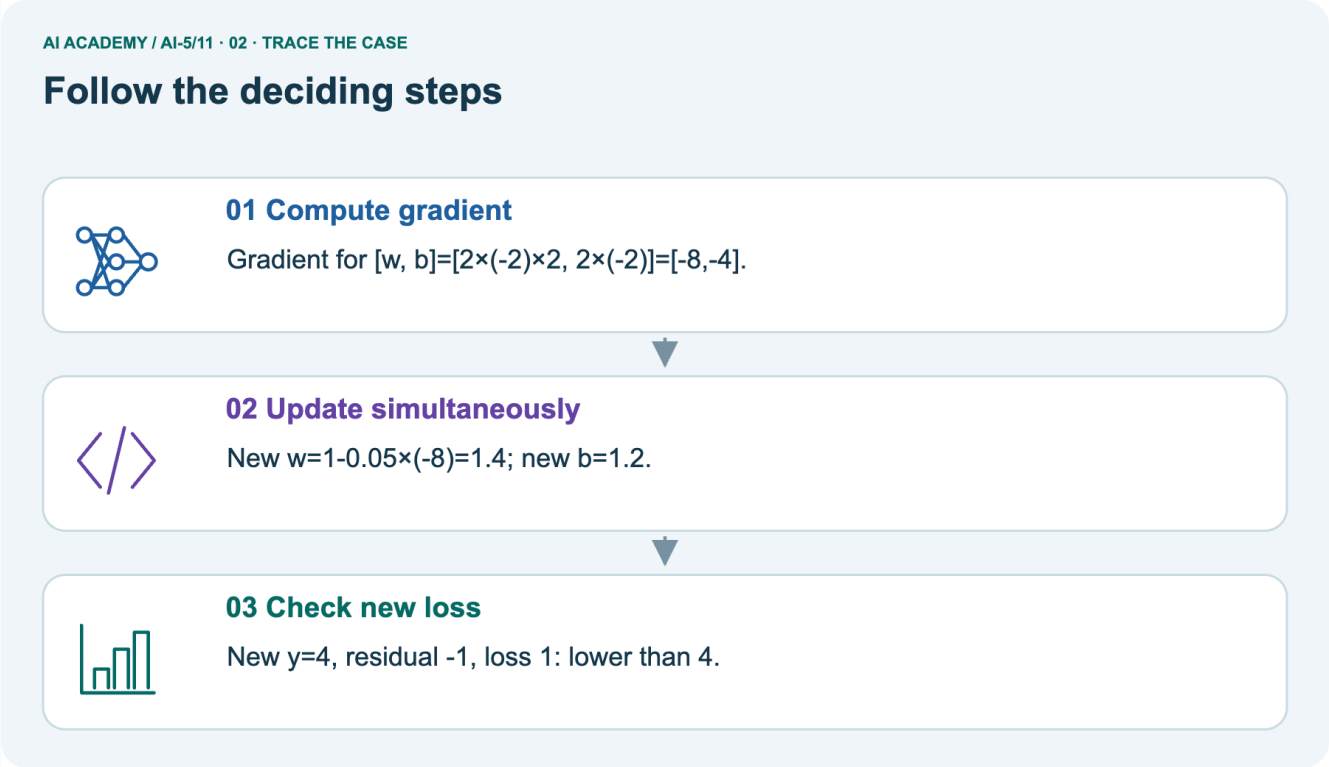
Follow the labelled picture with the matching explanation. Open the vector diagram

Use one explicit gradient update and then recompute the loss. Supplied facts: Model y=2w+b, target 5. Start w=1, b=1; **learning rate** 0.05. Initial y=3, error y−target=−2, squared loss 4.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

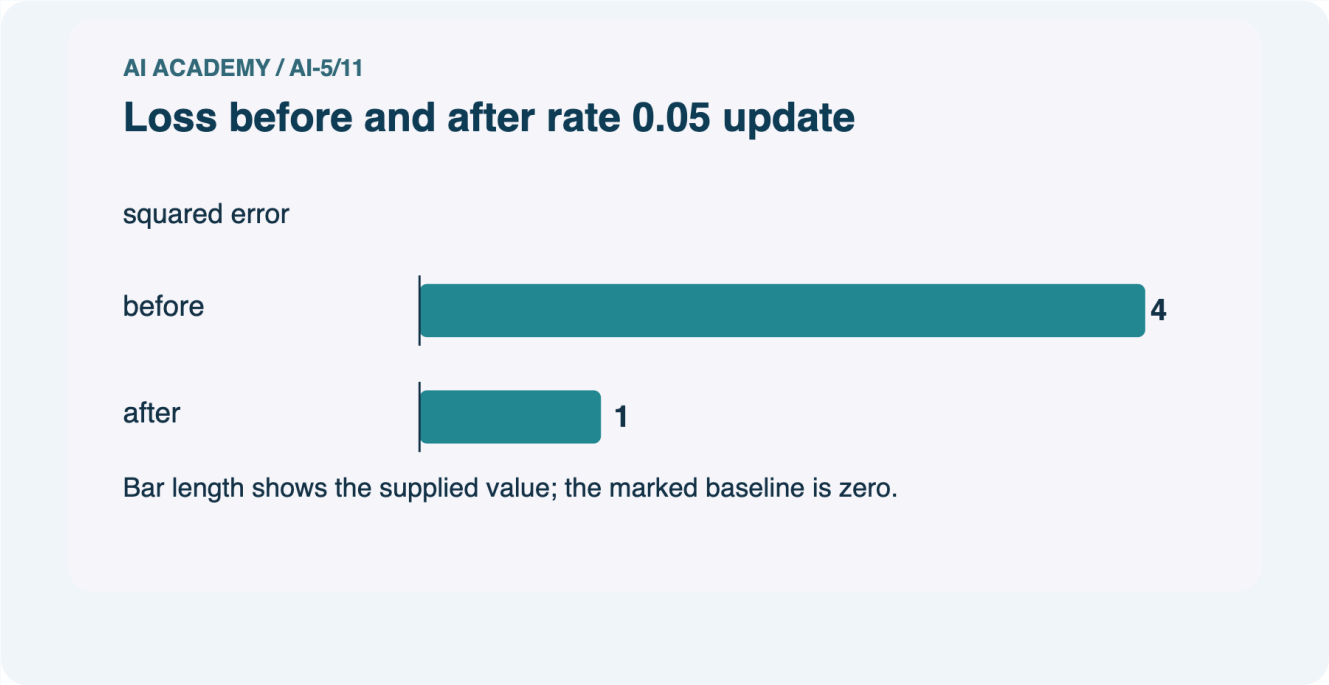


Follow the labelled picture with the matching explanation. Open the vector diagram

Compute gradient: Gradient for [w, b]=[2×(-2)×2, 2×(-2)]=[-8,-4]. → Update simultaneously: New w=1-0.05×(-8)=1.4; new b=1.2. → Check new loss: New y=4, residual -1, loss 1: lower than 4.

Point to each numbered stage and explain which recorded **input** it uses.

03. Loss before and after rate 0.05 update

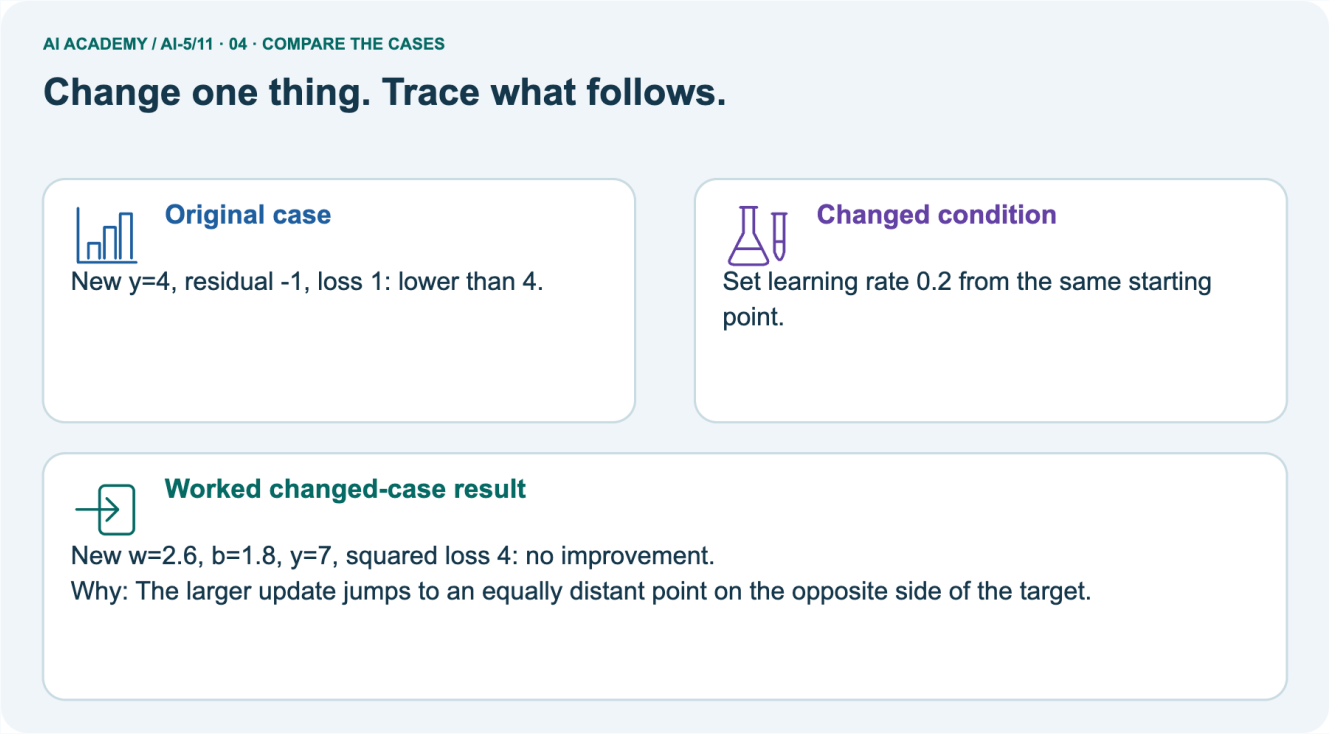


Follow the labelled picture with the matching explanation. Open the vector diagram

Loss before and after rate 0.05 update. Units: squared error. before: 4. after: 1.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



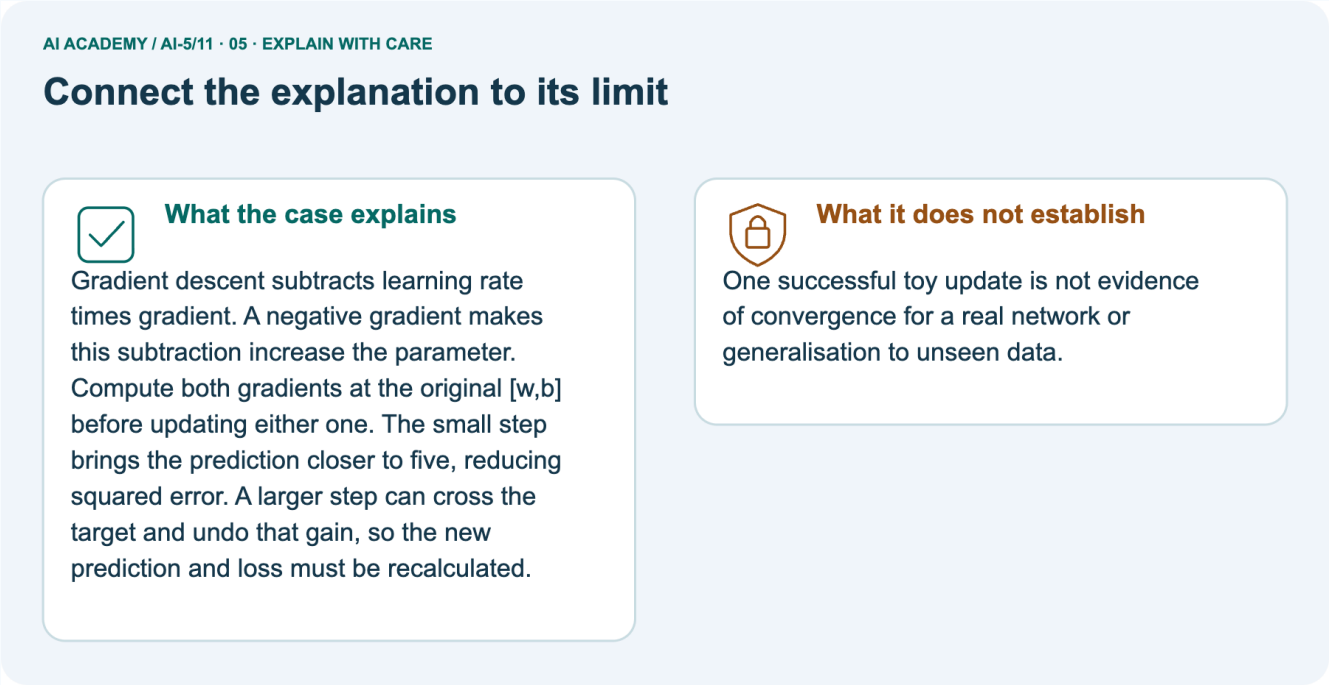
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: New y=4, residual -1, loss 1: lower than 4. Changed condition: Set **learning rate** 0.2 from the same starting point. Expected result: New w=2.6, b=1.8, y=7, squared loss 4: no improvement.

Reason: The larger update jumps to an equally distant point on the opposite side of the target.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

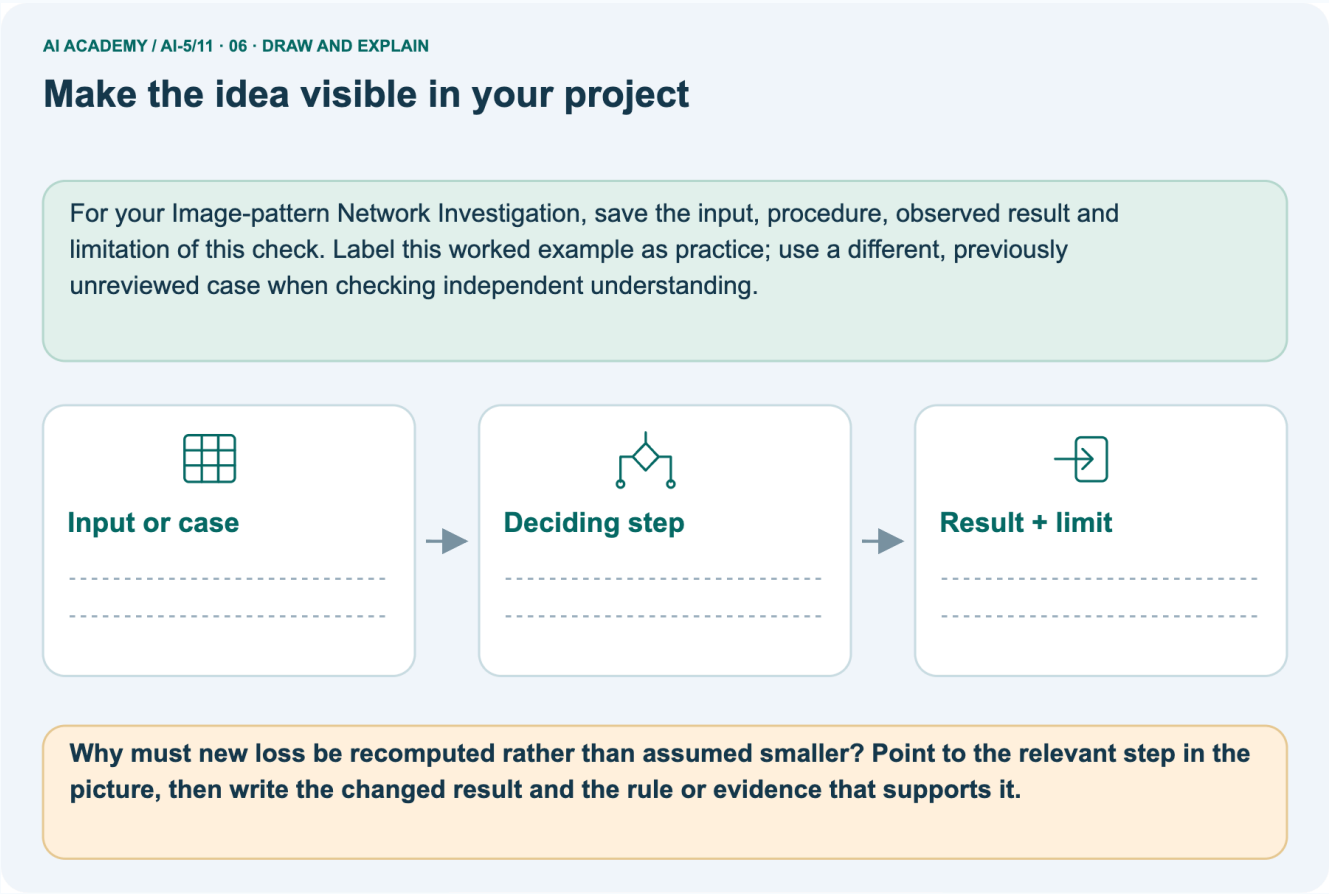
Gradient descent subtracts **learning rate** times gradient. A negative gradient makes this subtraction increase the parameter. Compute both gradients at the original [w,b] before updating either one.

The small step brings the **prediction** closer to five, reducing squared error. A larger step can cross the target and undo that gain, so the new **prediction** and loss must be recalculated.

Limit: One successful toy update is not **evidence** of convergence for a real network or generalisation to unseen data.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why must new loss be recomputed rather than assumed smaller? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.